

→ Formula :

1)  $n^{\text{th}}$  derivative

a)  $y = x^n$

$$y_n = n!$$

b)  $y = x^m$

$$y_n = m(m-1) \dots (m-n+1) x^{m-n}$$

c)  $y = \log(ax+b)$

$$y_n = (n-1)! (-1)^{n-1} a^n (ax+b)^{-n}$$

d)  $y = \frac{1}{ax+b}$

$$y_n = (-1)^n n! a^n (ax+b)^{-(n+1)}$$

e)  $y = \sin(ax+b)$

$$y_n = a^n \sin\left(\frac{n\pi}{2} + ax+b\right)$$

f)  $y = \cos(ax+b)$

$$y_n = a^n \cos\left(\frac{n\pi}{2} + ax+b\right)$$

2) Taylor's series

$$f(x) = f(a) + x f'(a) + \frac{x^2}{2!} f''(a) + \frac{x^3}{3!} f'''(a) + \dots$$

3) Maclaurin's series

$$f(x) = f(0) + x f'(0) + \frac{x^2 f''(0)}{2!}$$

4) Error

Max % error in  $x \rightarrow$  Taking all the

Relative error  $\rightarrow \frac{\Delta x}{x}$

$$\% \text{ error in } x = \frac{\Delta x}{x} \times 100$$

5) Test

a) P-series

$$V_n = \sum_{n=1}^{\infty} \frac{1}{n^p}$$

$p > 1 \rightarrow$  convergent

$p \leq 1 \rightarrow$  divergent

b) Comparison Test

$$\lim_{n \rightarrow \infty} \frac{\sum U_n}{\sum V_n} = \text{finite and non-zero}$$

then  $\sum U_n$  behaves as  $\sum V_n$

[n separate  $\sum$  are] ]

c) D' Alembert Ratio's test

$$\lim_{n \rightarrow \infty} \frac{U_n}{U_{n+1}} = L \text{ (say)}$$

$L > 1 \rightarrow$  convergent

$L < 1 \rightarrow$  divergent

$L = 1 \rightarrow$  Test Fail

d) Cauchy Root test

$$\lim_{n \rightarrow \infty} U_n^{1/n} = L$$

$L > 1 \rightarrow$  divergent

$L < 1 \rightarrow$  convergent

$L = 1 \rightarrow$  Test fail

Power

में  $n$

हो

e) Raabe's Test

$$\lim_{n \rightarrow \infty} n \left( \frac{U_n}{U_{n+1}} - 1 \right) = L \text{ (say)}$$

$L > 1 \rightarrow$  convergent

$L < 1 \rightarrow$  divergent

$L = 1 \rightarrow$  Test fail

( $n$  separate नहीं हो)



### f) log Test

$$\lim_{n \rightarrow \infty} n \log \frac{U_n}{U_{n+1}} = L$$

$L > 1 \rightarrow$  Convergent

$L < 1 \rightarrow$  Divergent

$L = 1 \rightarrow$  Fail

किसी Term  
की power  
में  $n$  हो  
और किसी  
में नहीं

### g) Alternating Series

$\rightarrow$  Absolutely  $\rightarrow$  both are convergent  
[  $\sum |U_n|$  and  $\sum U_n$  ]

$\rightarrow$  Conditionally  $\rightarrow$  both are opposite

### h) Roc

#### a) Cartesian form

$$f = \frac{[1 + y_1^2]^{3/2}}{y_2}$$

#### b) Polar Curve

$$f = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - r r_2}$$

- Parametric coordinates of
- a) PB  $\rightarrow$  at  $t$ ,  $2at \rightarrow y^2 = 4ax$
  - b) Ellipse  $\rightarrow (x = a \cos \theta, y = b \sin \theta)$
  - c) HB  $\rightarrow x = a \sec \theta, y = b \tan \theta$

c) Parametric Curve

$$f = \frac{[(x')^2 + (y')^2]^{\frac{3}{2}}}{x'y'' + y'x''}$$

1) Asymptotes

T-1)  $\parallel$  to  $x$ -axis

Put coefficient of highest power of  $x$  equal to zero

T-2)  $\parallel$  to  $y$ -axis

Put coefficient of highest power of  $y$  equal to zero

T-3: oblique asymptote

Put  $x=1$  and  $y=m$  in highest degree (we get  $\phi_n(m)$ )

$$\text{Put } \phi_n(m) = 0$$

a)  $m$  has distinct roots

$$c = - \frac{\phi_{n-1}(m)}{\phi_n'(m)}$$

b)  $m$  has repeated roots

for 2 roots

$$\frac{c^2}{2} \phi_n''(m) + \phi_{n-1}'(m) c + c^0 \phi_{n-2}(m) = 0$$

for 3 roots

$$\frac{c^3}{3} \phi_n'''(m) + \frac{c^2}{2} \phi_{n-1}''(m) + c \phi_{n-2}'(m) + \phi_{n-3}(m) = 0$$

8) Gamma functions

$$\Gamma n = \int_0^{\infty} e^{-x} x^{n-1} dx, \quad \Gamma \frac{1}{2} = \sqrt{\pi}$$

$$\frac{\sqrt{n+1}}{n \notin \mathbb{N}} \quad | \quad n \in \mathbb{N} \quad \frac{1}{n!}$$

$n\Gamma n \qquad \qquad \qquad n!$

9) Beta functions

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$\beta(m, n) = \frac{\Gamma n \Gamma m}{\Gamma m+n}$$

10) Formula



$$(i) \int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{\sqrt{\frac{p+1}{2}} + \sqrt{\frac{q+1}{2}}}{2\sqrt{\frac{p+q+2}{2}}}$$

### Unit - 3

1) Hermitian Matrix:  $A^\dagger = A$   
 $A^\dagger = (\bar{A})' / (A')$

2) Skew Hermitian Matrix:  
 $A = -A^\dagger$

3) Unitary Matrix:  $AA^\dagger = I$

4) Transformation,

|              | Row | Column |
|--------------|-----|--------|
| Gauss Jordan | ✓   | X      |
| Normal form  | ✓   | ✓      |
| Echelon      | ✓   | X      |

5) Inverse (Gauss Jordan)

$$\begin{array}{ccc} A & \sim & I A \\ \downarrow & & \downarrow \\ I & \rightsquigarrow & A^{-1} \end{array}$$

6) RANK  $[J(A)]$

(i) Normal: Convert  $A$  into  $I$

$$J(A) = \text{Order of } I$$

(ii) Echelon: make  $A \rightarrow$  Upper  $\Delta$  matrix

7) Linear Dependence / Independence

$\therefore J(A) < \text{no. of vectors} \rightarrow \text{dependent}$

$\therefore J(A) = \text{no. of vectors} \Rightarrow \text{independent}$

8) Solution of Linear equations

[Augmented matrix  $[A:B]$ ]

i) Homogeneous  
 $A\mathbf{x} = \mathbf{0}$

Non-homogeneous  
 $A\mathbf{x} = \mathbf{B}$



→ For Non-Homogeneous

a) if  $\rho(A) = \rho(A:B)$

eq. are consistent and solution is possible

b) if  $\rho(A) = \rho(A:B) = \text{no. of variables}$   
→ Solution is Unique.

c) if  $\rho(A) = \rho(A:B) < \text{no. of variables}$   
Solution is many / many solutions

9) Cayley-Hamilton Theorem  
Put  $|A - \lambda I| = 0$  for  
characteristic equation.

10) Eigen values and vectors

Find value of  $\lambda$  from  $|A - \lambda I| = 0$

→ Find eigen <sup>vectors</sup> ~~value~~ of each  $\lambda$

by  $[A - \lambda I] x_1 = 0$

where  $x_1$  is eigen ~~value~~ vector

[value of  $\lambda$  is eigen value]

## 11) Diagonalisation Matrix

$$D = P^{-1}AP, \quad P \rightarrow \text{Modal Matrix}$$
$$A = PDP^{-1}$$

→ In Diagonal matrix, Diagonal value  
i) equal to  $-1$

$$P^{-1} = \frac{\text{adj } P}{|P|}$$

## 12) Quadratic form.

$$A = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \quad \text{QF: } x^T A x$$

$$x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad ax^2 + by^2 + cz^2 + 2hxy + 2gzx + 2fyz$$

→ Reduction to CF

$$\text{CF: } \lambda_1 x^2 + \lambda_2 y^2 + \lambda_3 z^2$$